

Investigating sequence-to-splicing prediction from personal genomes

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Mid-Atlantic Splicing Conference

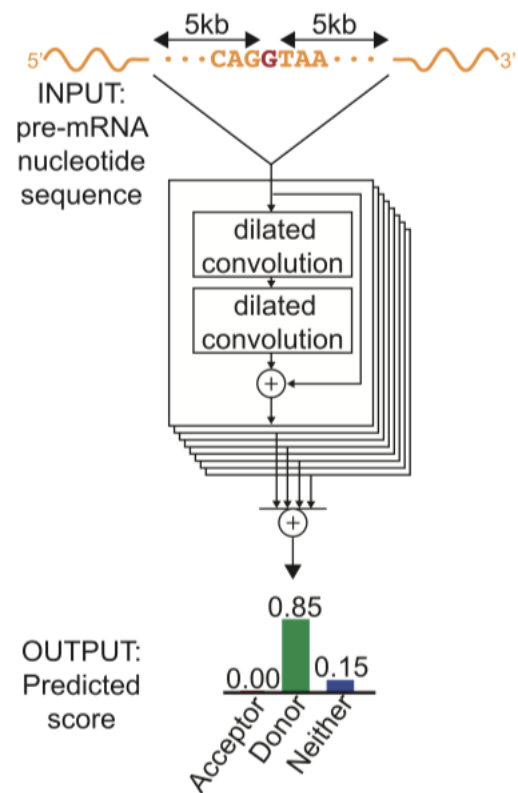
September 9th, 2025



Overview

- Review of current deep learning splicing models.
- Training deep learning splicing models from paired personal genomes & transcriptomes.
- Model ablations on predicting splice sites and usage across individuals & tissues.
- Variant effect prediction from MPRA.
- Variant effect prediction from RBP specific dataset.

SpliceAI

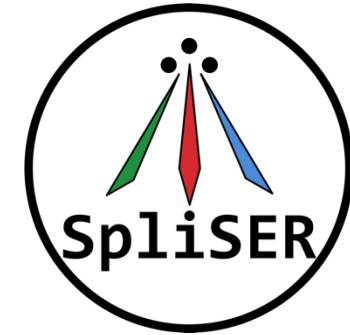
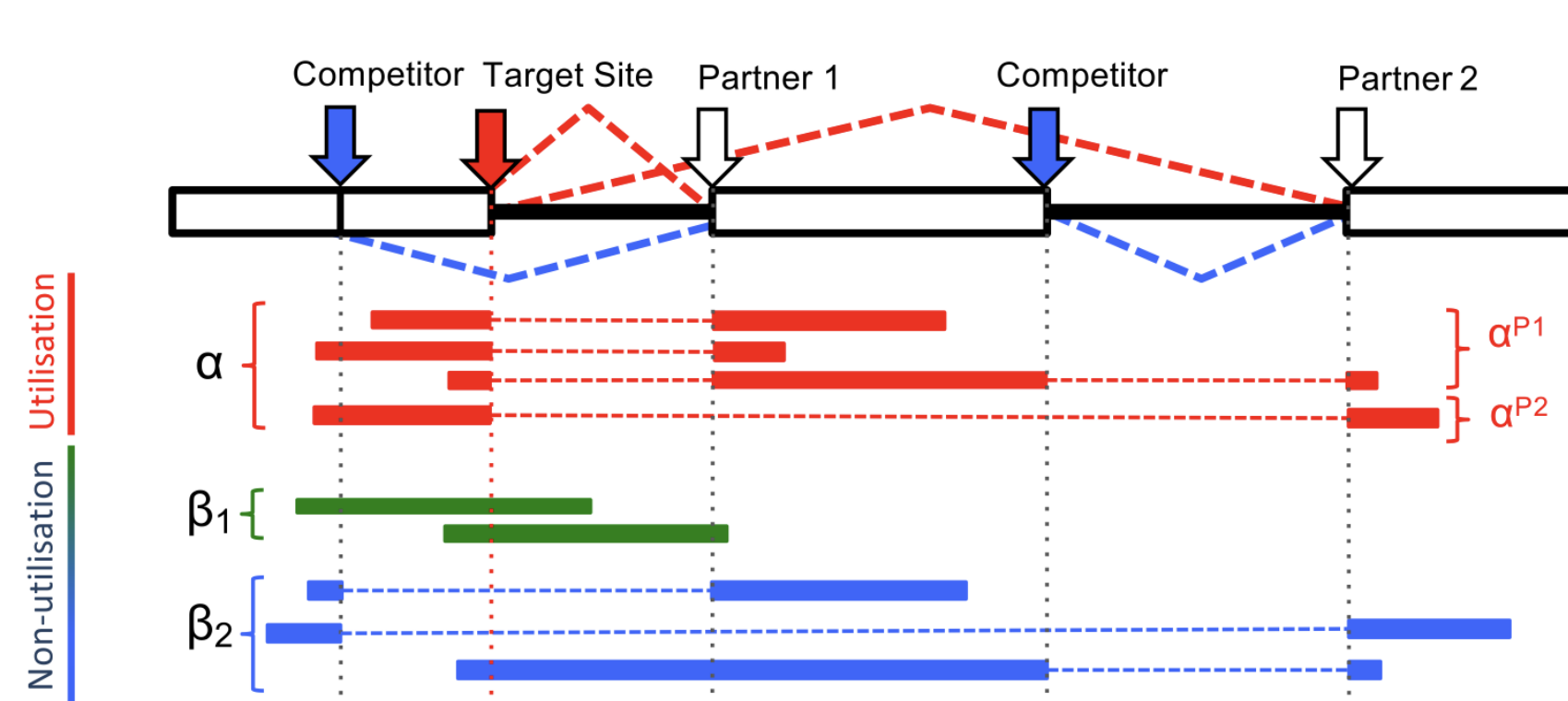


	Top-k accuracy	PR-AUC
SpliceAI-80nt	0.57	0.60
SpliceAI-400nt	0.90	0.95
SpliceAI-2k	0.93	0.97
SpliceAI-10k	0.95	0.98
GeneSplicer	0.30	0.23
MaxEntScan	0.22	0.15
NNSplice	0.22	0.15

- First model to achieve high accuracy for detecting splice sites and splice disruptive variants.
- First “uninformed” splicing model.
- 10kb Pre-mRNA sequence used as input to predict the probability of a base being: acceptor, donor, or neither.
- Data: GENCODE junctions of primary transcripts.

Jaganathan et al., Cell, 2019.

Splice Site Usage (SSU) is PSI at single base resolution



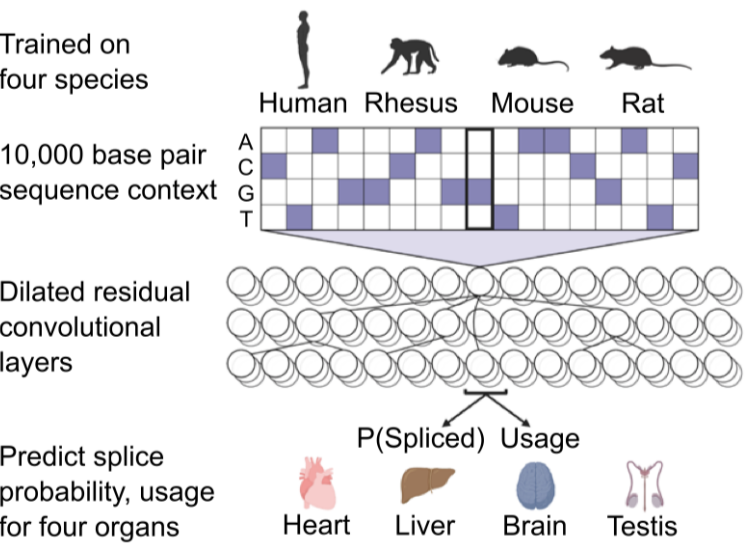
$$SSU = \frac{\alpha}{\alpha + \beta_1 + \beta_2}$$

$$0 \leq SSU \leq 1$$

Dent et al., *NAR Genomics and Bioinformatics*, 2021.

Pangolin addressed tissue specificity and Splice Site Usage (SSU)

a Architecture and training

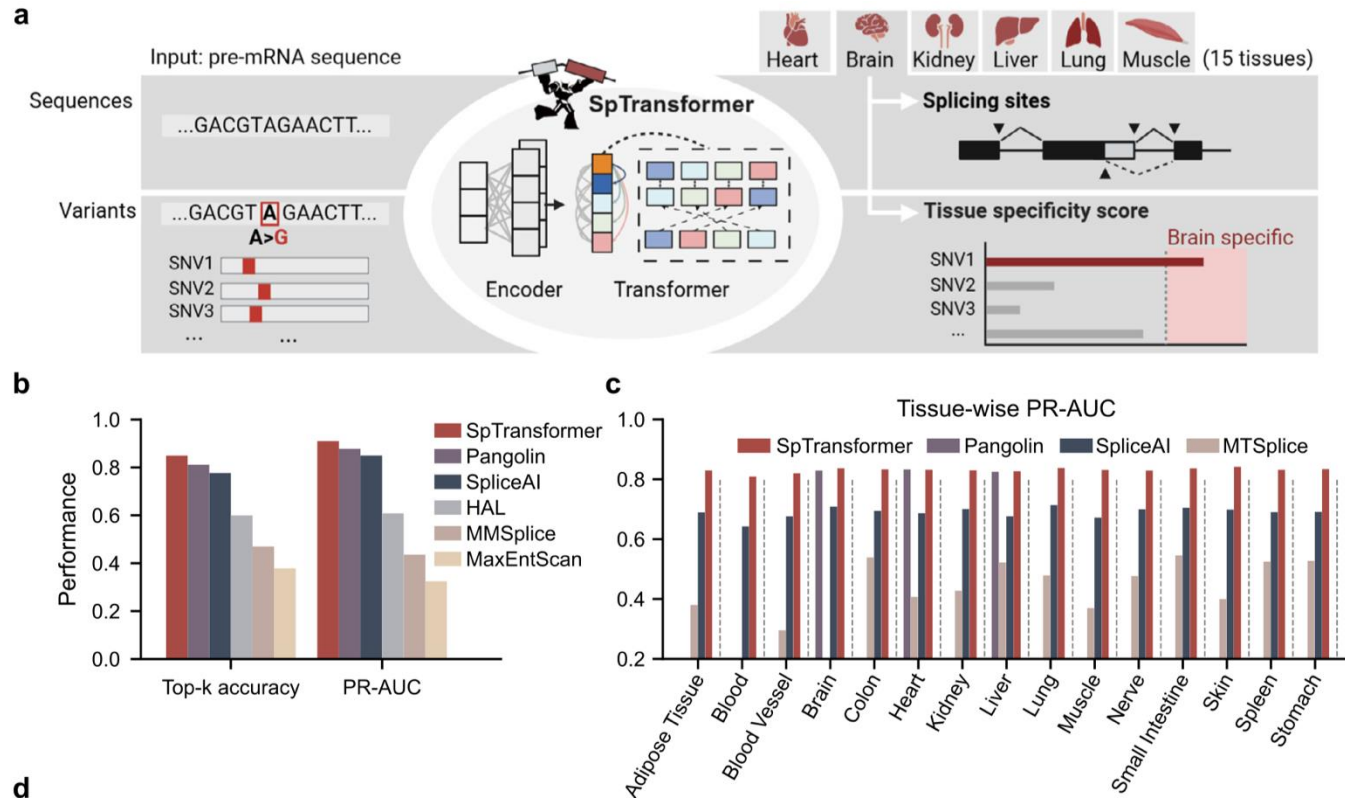


b Pangolin outperforms existing methods on splice site prediction

	Top-1					Top-0.5					AUPRC				
Heart	83	81	34	23	17	95	88	40	28	18	88	82	27	16	10
Liver	77	73	33	23	18	92	79	40	27	20	83	71	27	15	11
Brain	80	77	38	26	20	95	89	46	32	23	86	79	32	19	12
Testis	74	68	41	28	24	94	92	50	35	28	81	74	35	21	16
	Pangolin	SpliceAI	HAL	MMSplice	MaxEntScan	Pangolin	SpliceAI	HAL	MMSplice	MaxEntScan	Pangolin	SpliceAI	HAL	MMSplice	MaxEntScan

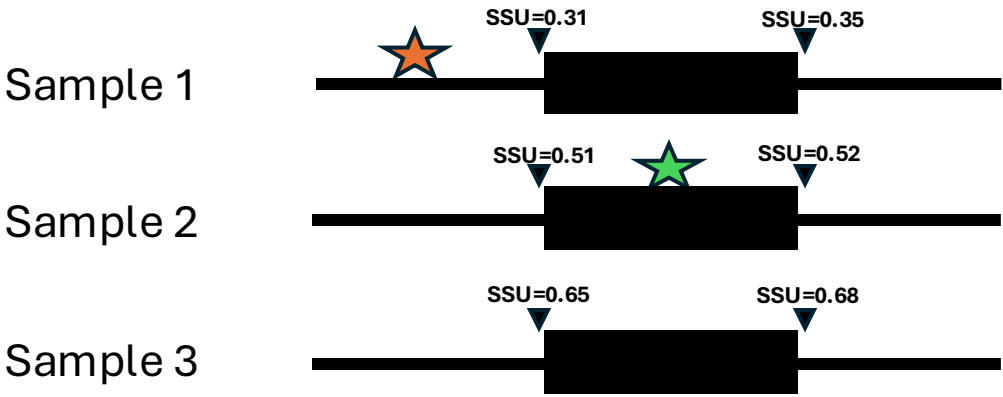
- Same architecture as SpliceAI with additional output head to predict SSU
- Data: RNA-Seq from four tissues and four species.
 - Identify location of splice sites
 - Quantify SSU
- Predicts the probability of a base being a splice site and usage in four tissues.

SpliceTransformer increased tissues and altered architecture

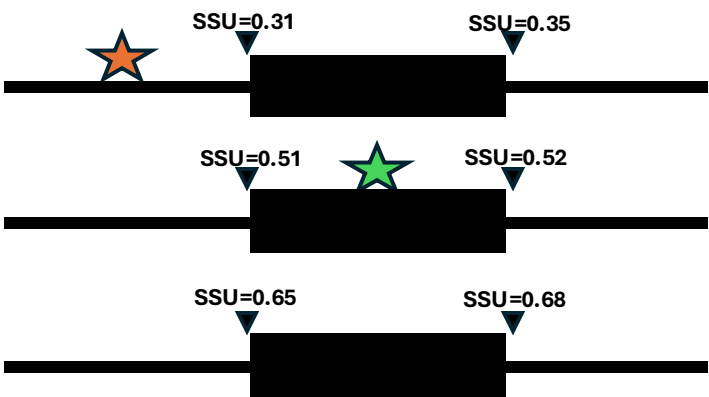


- SpliceAI encoder followed by sinkhorn transformer.
- Data: GTEx exon-exon junctions of 15 tissues.
- SSU definition: the proportion of samples in which the splice site was detected.
- Predicts acceptor, donor, neither probabilities and usage in 15 tissues.

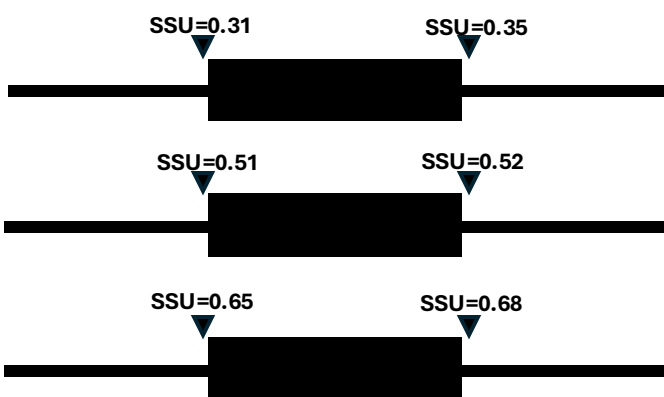
Generation of paired genome/transcriptome dataset



Variant dataset



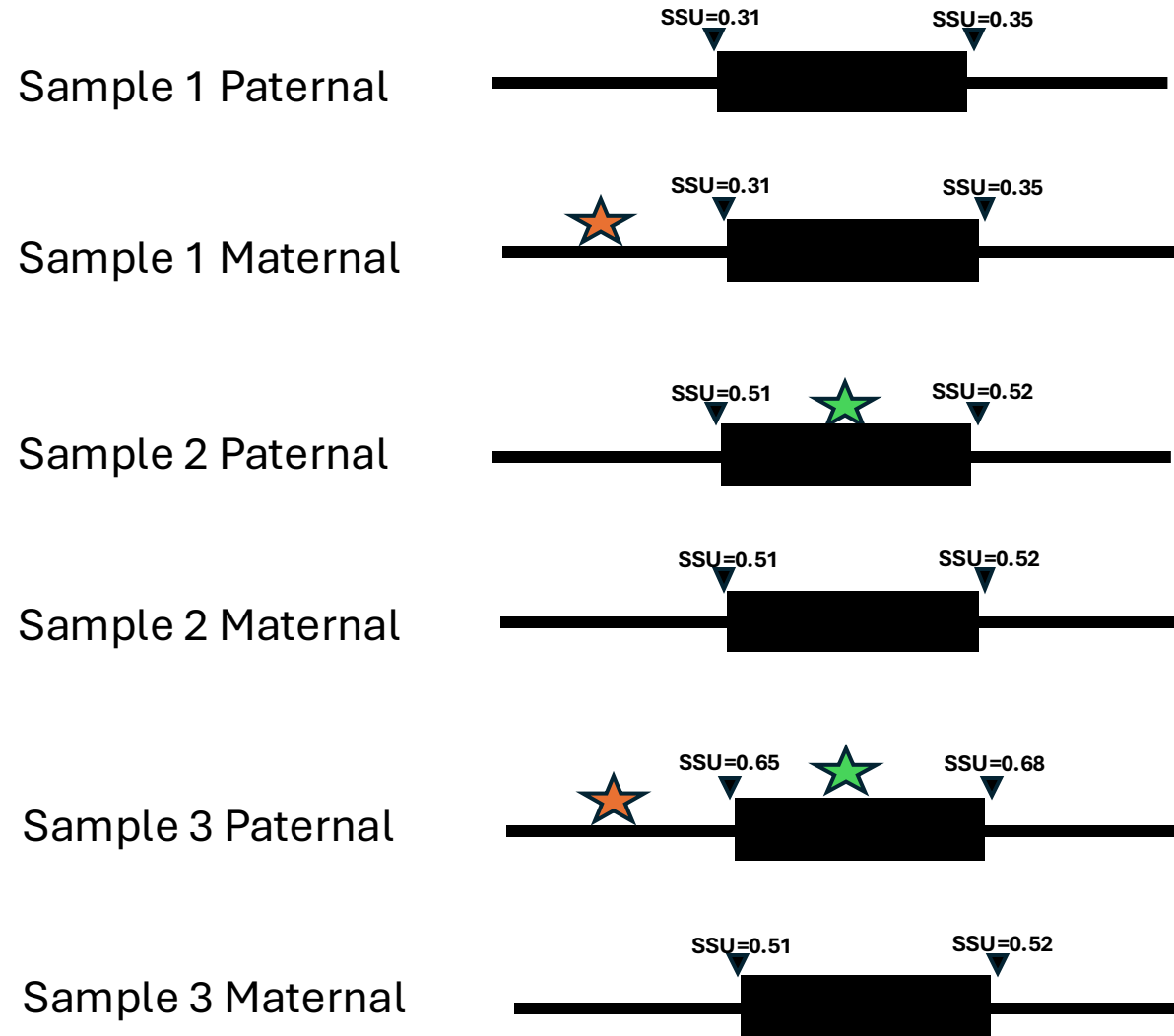
Reference dataset



SpliceAI, Pangolin, & SpliceTransformer dataset



Limitation of paired genome/transcriptome dataset

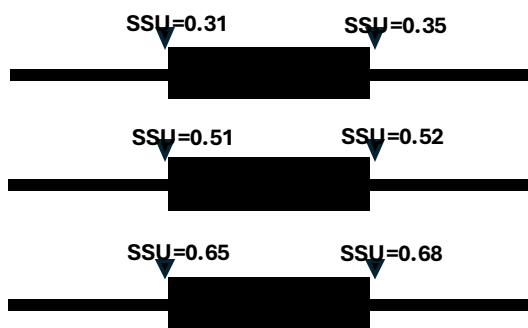


Model training with reference and personal genomes

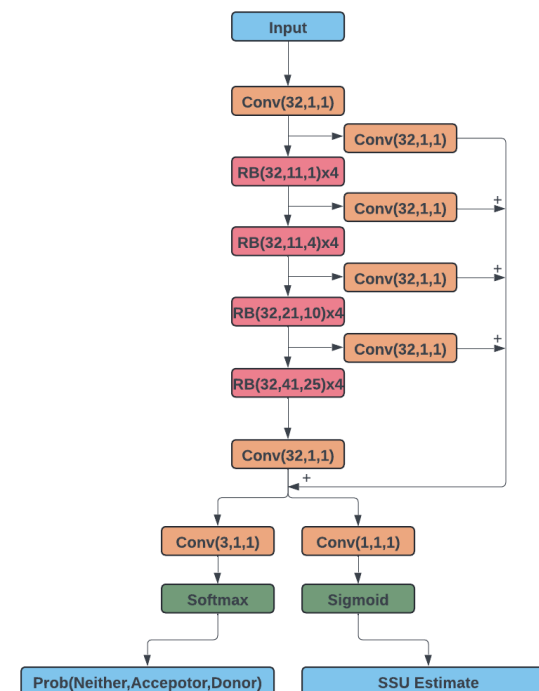
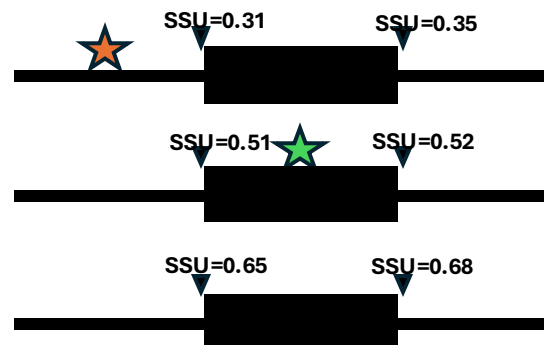
Train

- 100 individuals HAEC (human airway epithelial cell)
- Chromosomes 2,4,6,8, 9-22
 - Protein-coding genes

DCNN-reference



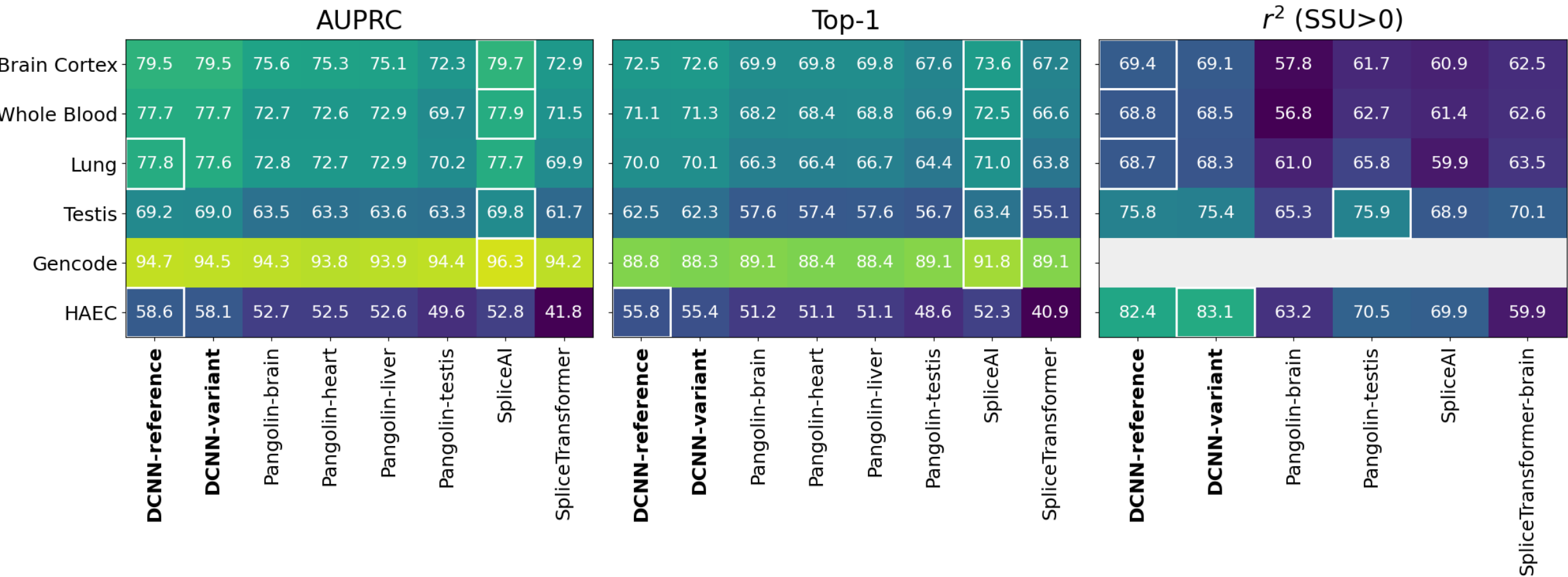
DCNN-variant



Test

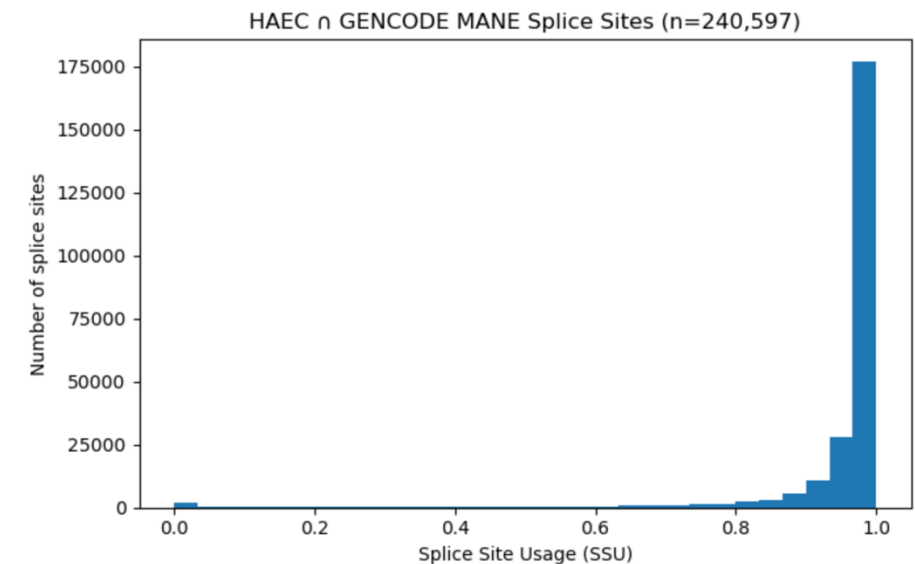
- Chromosomes 1,3,5,7
 - Non-paralogous genes
 - Protein-coding
- 82 held out HAEC individuals
- 10 held out GTEx individuals
 - Lung
 - Testis
 - Brain Cortex
 - Whole Blood
- GENCODE splice sites of primary transcripts

Heatmap summarizing classification & regression performance



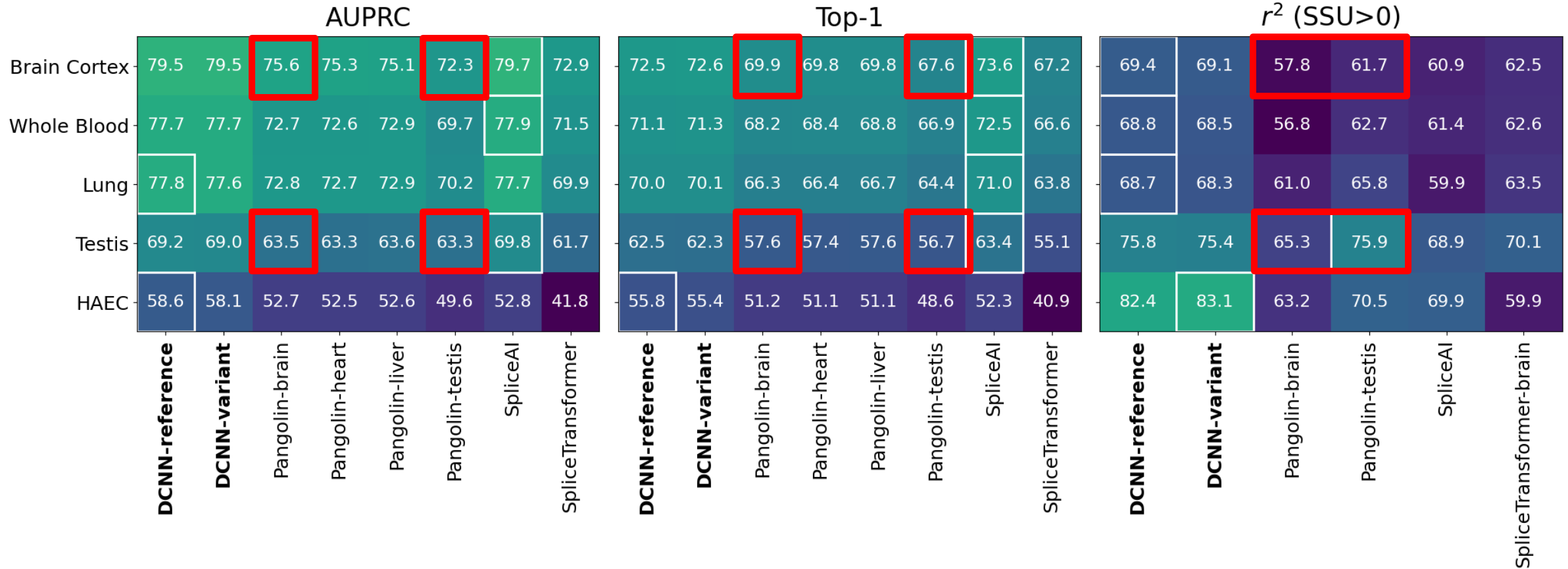
GENCODE classification task is biased

	AUPRC								Top-1							
	DCNN-reference	DCNN-variant	Pangolin-brain	Pangolin-heart	Pangolin-liver	Pangolin-testis	SpliceAI	SpliceTransformer	DCNN-reference	DCNN-variant	Pangolin-brain	Pangolin-heart	Pangolin-liver	Pangolin-testis	SpliceAI	SpliceTransformer
Brain Cortex	79.5	79.5	75.6	75.3	75.1	72.3	79.7	72.9	72.5	72.6	69.9	69.8	69.8	67.6	73.6	67.2
Whole Blood	77.7	77.7	72.7	72.6	72.9	69.7	77.9	71.5	71.1	71.3	68.2	68.4	68.8	66.9	72.5	66.6
Lung	77.8	77.6	72.8	72.7	72.9	70.2	77.7	69.9	70.0	70.1	66.3	66.4	66.7	64.4	71.0	63.8
Testis	69.2	69.0	63.5	63.3	63.6	63.3	69.8	61.7	62.5	62.3	57.6	57.4	57.6	56.7	63.4	55.1
Gencode	94.7	94.5	94.3	93.8	93.9	94.4	96.3	94.2	88.8	88.3	89.1	88.4	88.4	89.1	91.8	89.1
HAEC	58.6	58.1	52.7	52.5	52.6	49.6	52.8	41.8	55.8	55.4	51.2	51.1	51.1	48.6	52.3	40.9

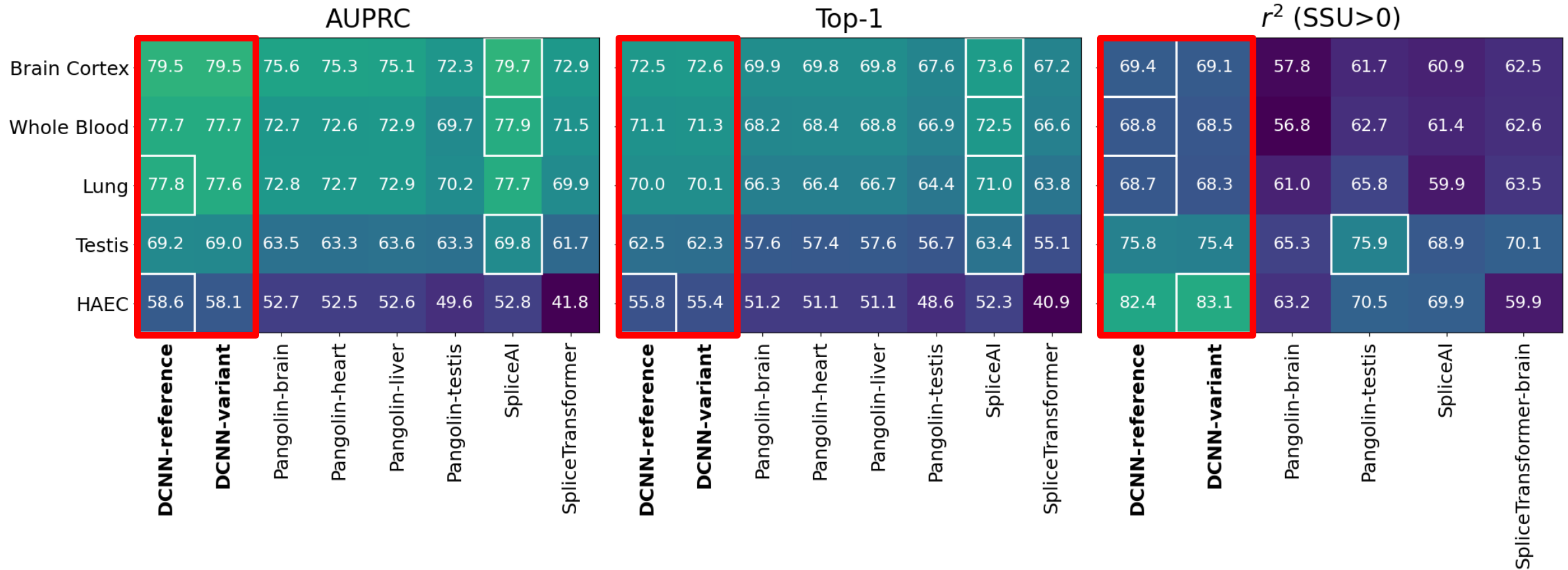


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Tissue specific models lack specificity



No gains from training with personal genomes



Benchmarking of deep neural networks for predicting personal gene expression from DNA sequence highlights shortcomings

[Alexander Sasse](#), [Bernard Ng](#), [Anna E. Spiro](#), [Shinya Tasaki](#), [David A. Bennett](#), [Christopher Gaiteri](#), [Philip](#)

[L. De Jager](#), [Maria Chikina](#) ✉ & [Sara Mostafavi](#) ✉

[Nature Genetics](#) **55**, 2060–2064 (2023) | [Cite this article](#)

Fine-tuning sequence-to-expression models on personal genome and transcriptome data

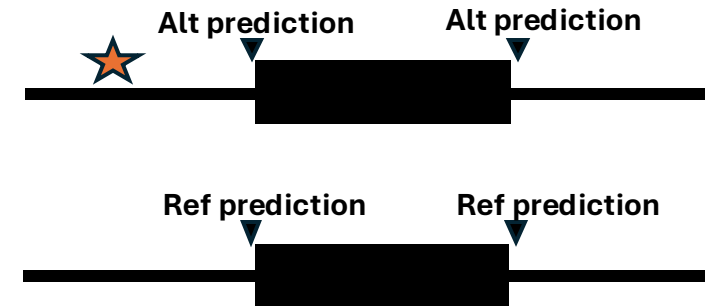
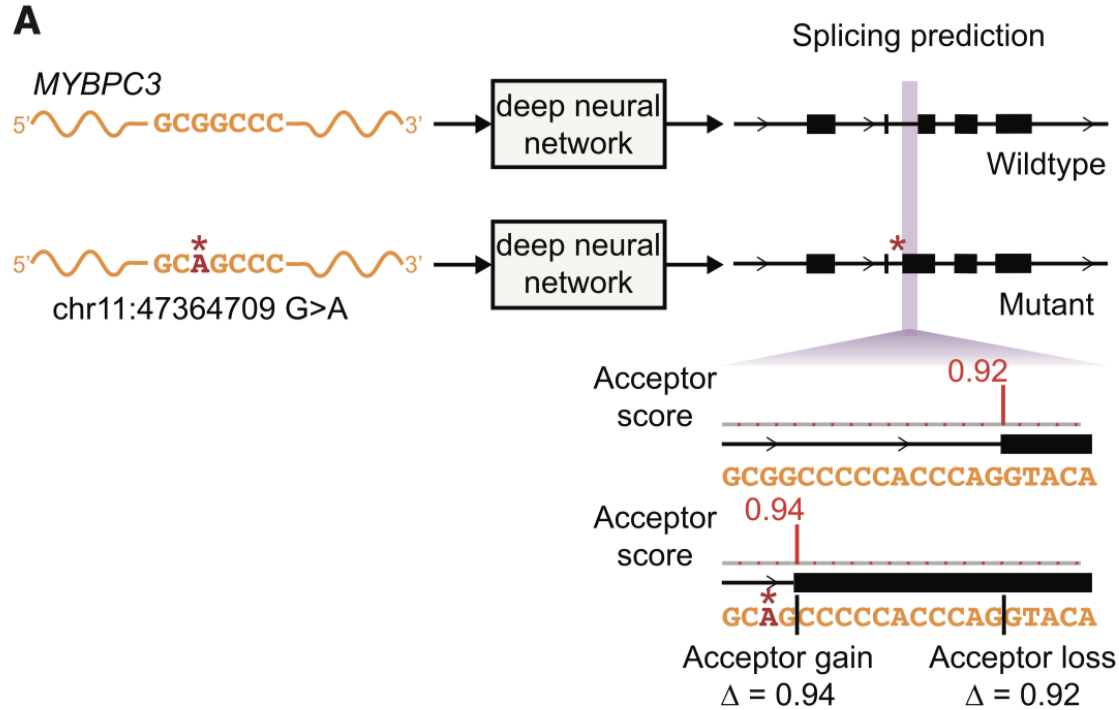
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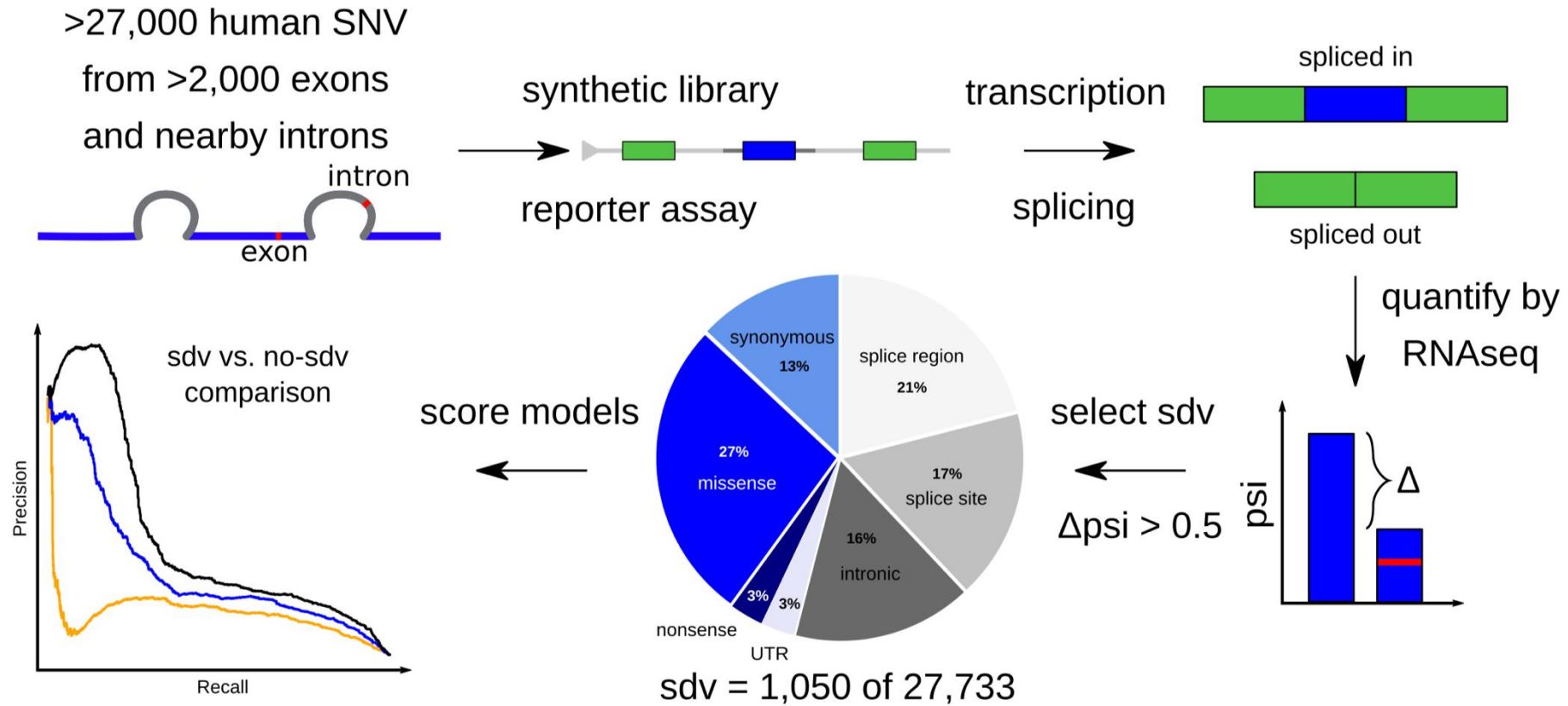
Delta scores for variant effect prediction



$$\text{PSI} = \text{mean}(\text{acceptor_score}, \text{donor_score})$$

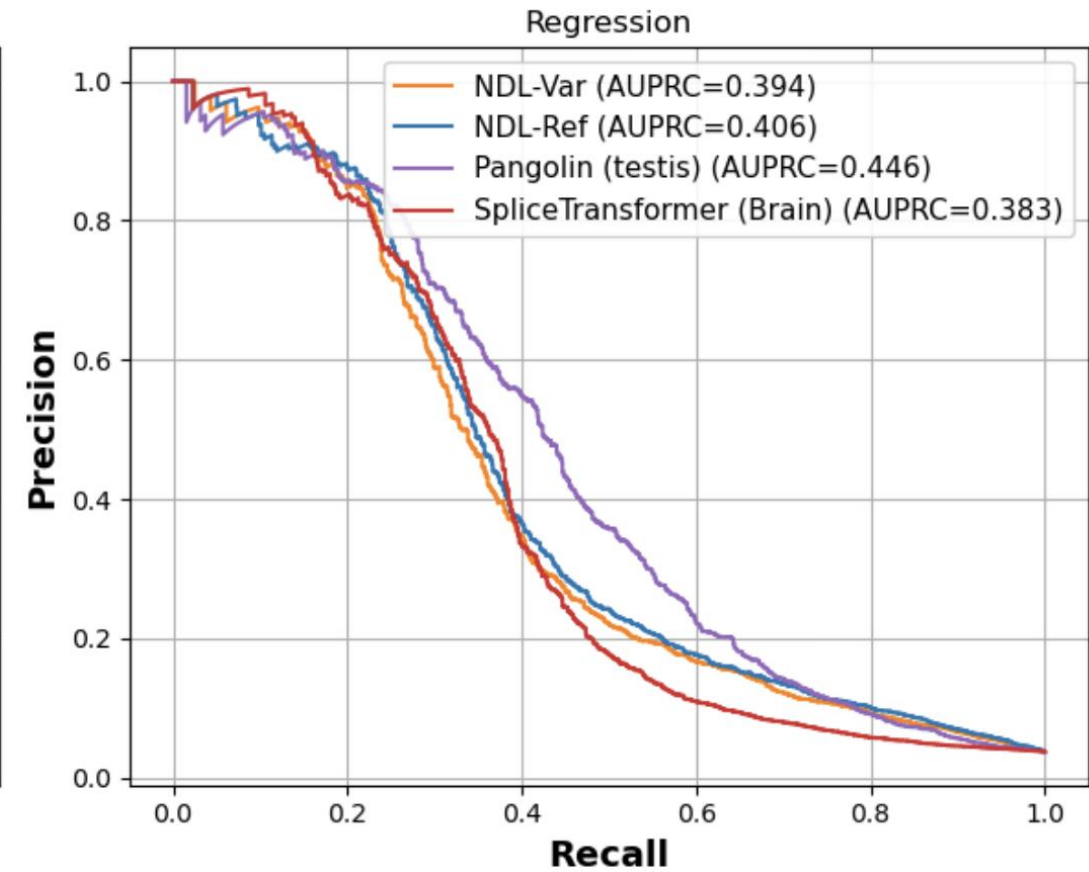
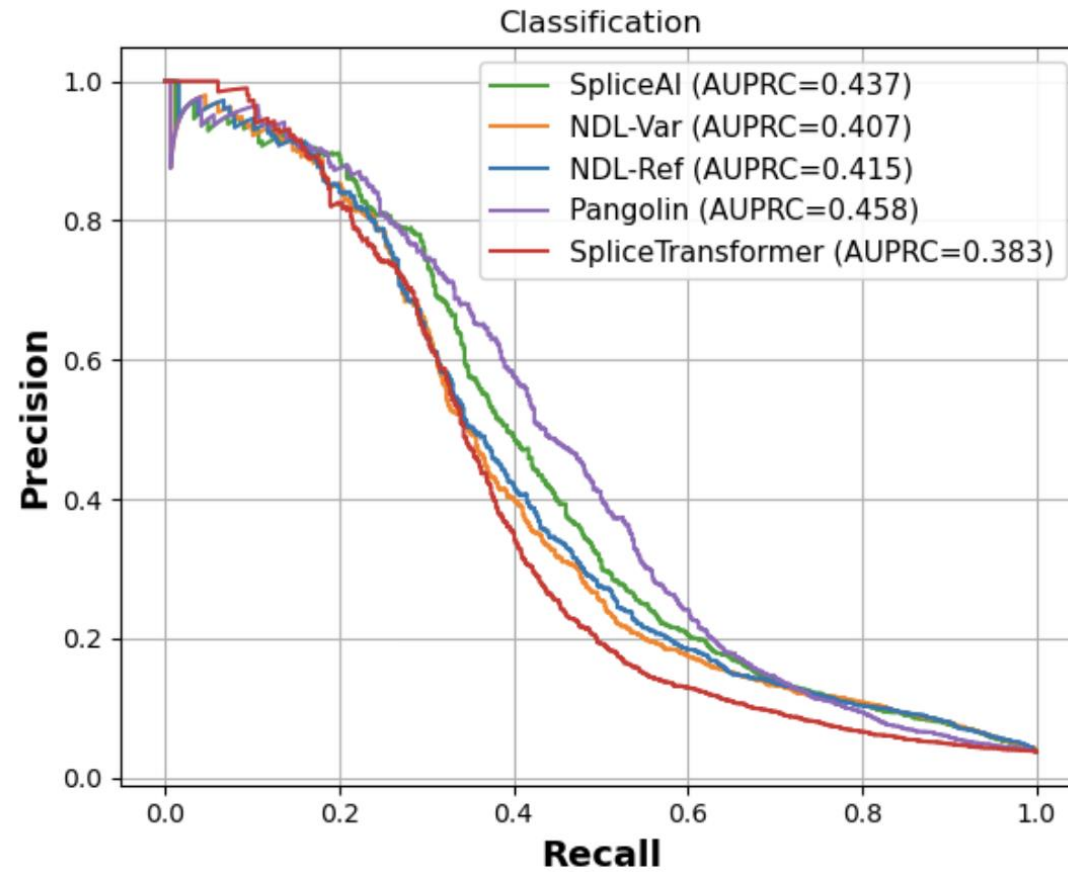
$$\Delta\text{PSI} = \text{PSI_Alt} - \text{PSI_Ref}$$

Benchmarking predictors on the MFASS dataset

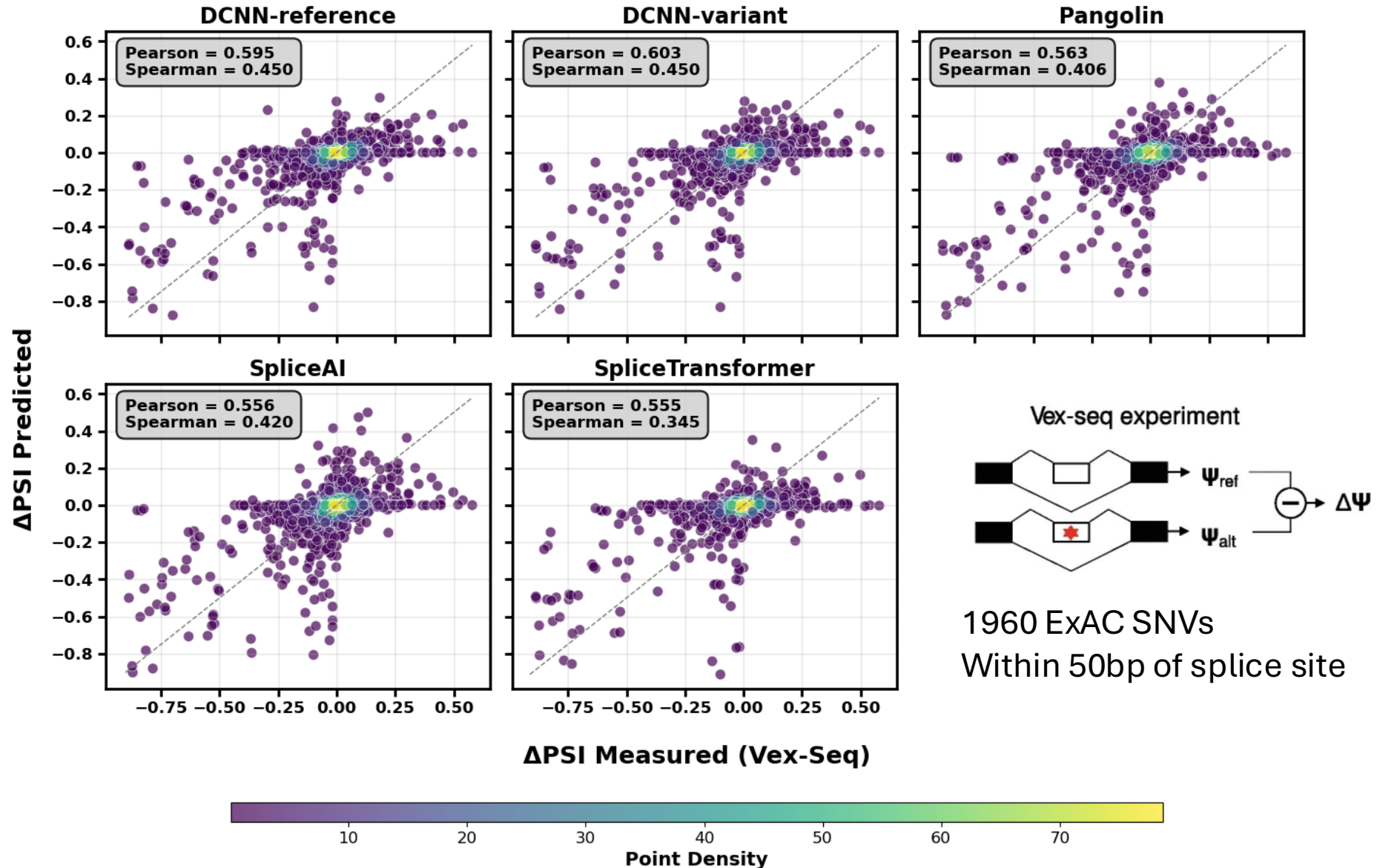


Rentzsch *et al.*, *Genome Medicine*, 2021.

Benchmarking predictors on the MFASS dataset



Benchmarking predictors on the Vex-Seq dataset

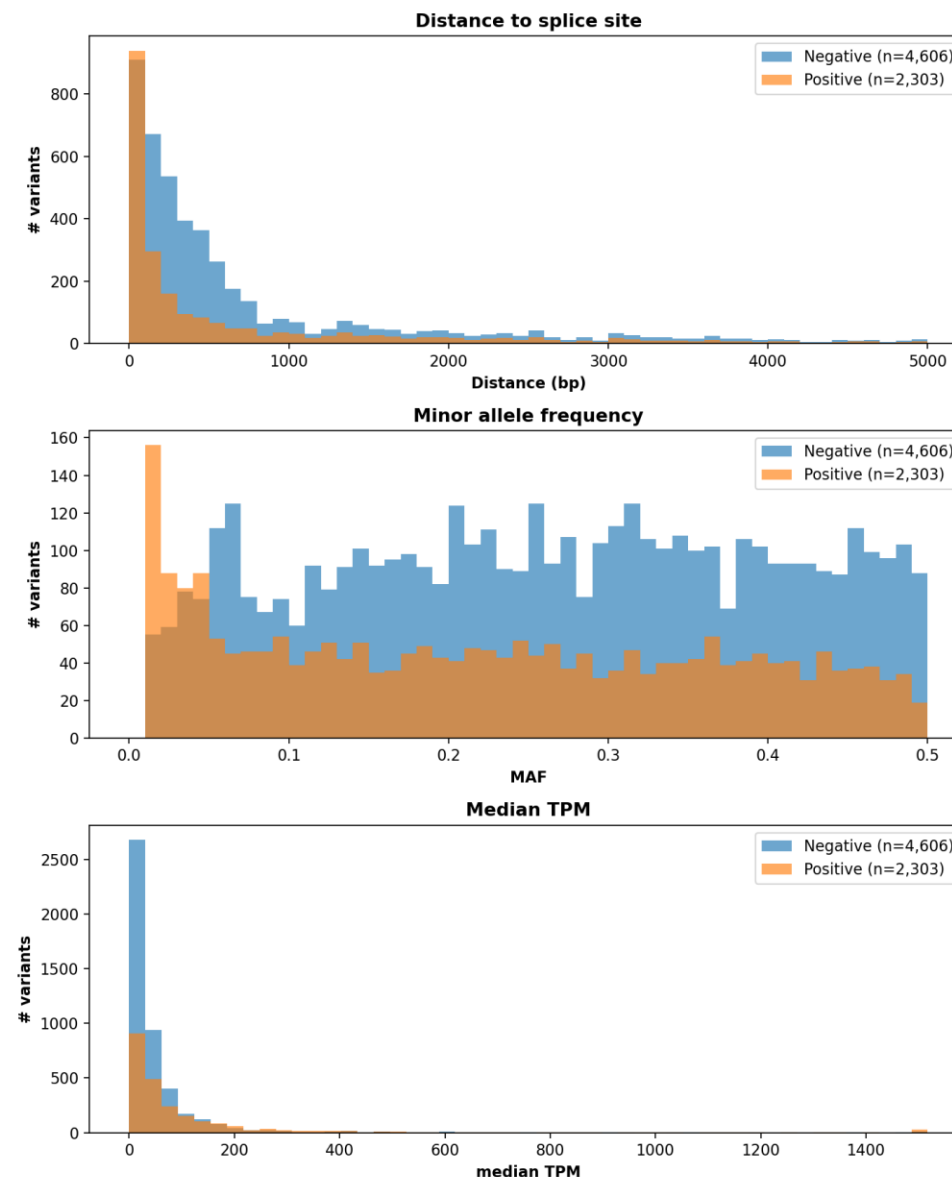


Generating an RBP specific variant dataset

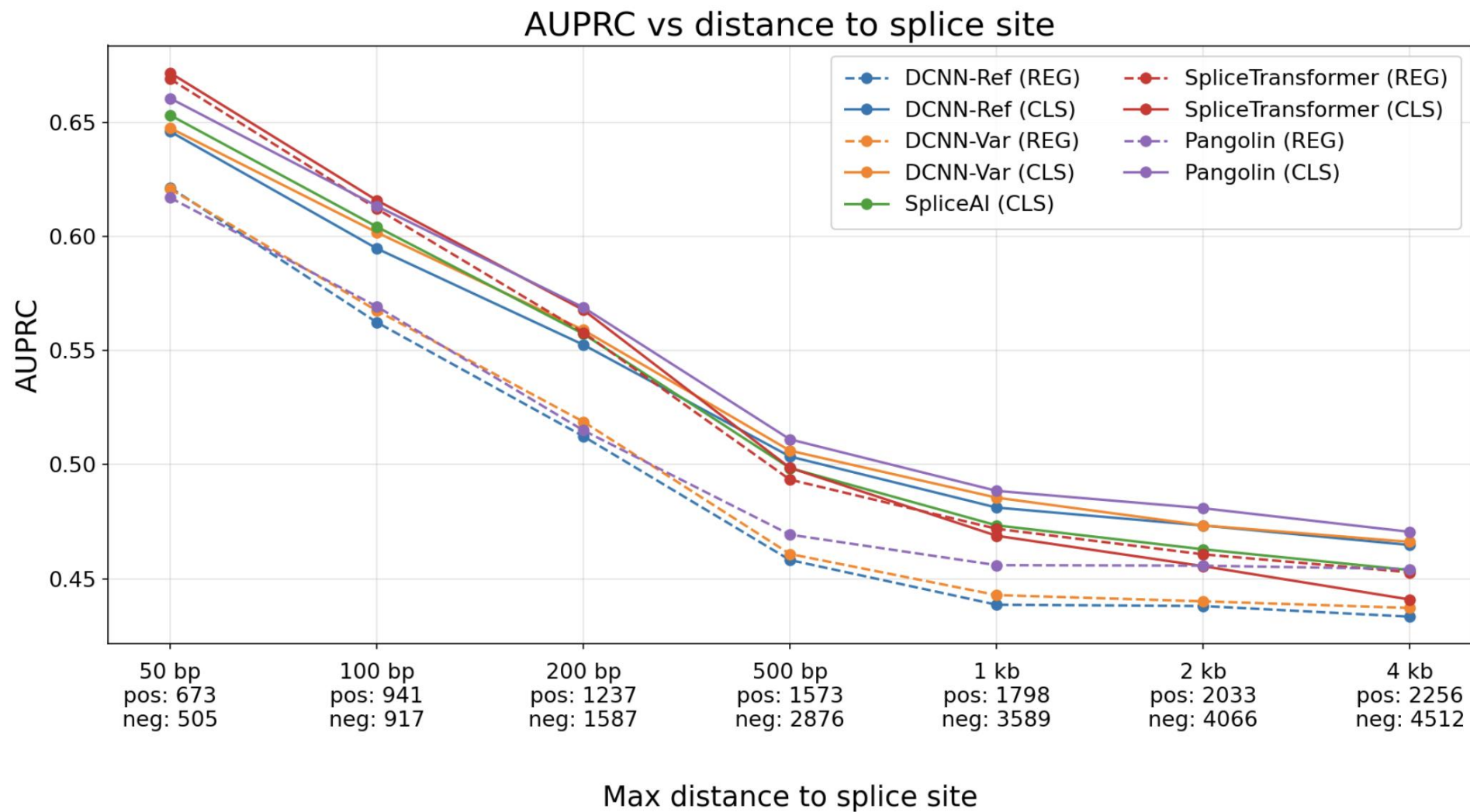


- Leafcutter fine-mapped sQTLs from 49 GTEx tissues.
 - Within 5kb of splice site

- Motifs of 159 RBPs from multiple databases.
- Scan sequences around each variant for matches to each motif (p -value $< 1e-4$).
- Retain variants with overlapping motif hit.
- Create sets of casual and non-casual variants
 - Positive set: PIP > 0.9
 - Negative set: PIP < 0.01
- Match positive (casual) variants to negatives based on distance and MAF.



Generating an RBP specific variant dataset



Conclusions & future directions

- Training with personal genomes offers no gains.
 - Develop effective methods to fine-tune models on splice disruptive variants.
- The degree of model tissue specificity is unclear and understudied.
 - More extensive interpretability analysis focused on RBPs and tissue specific splice sites.
- What are optimal approaches for generating and leveraging pre-trained RNA embeddings?

Thank you!

Team:

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